

Aaron Low

TECHNICAL ARTIST · ANIMATION TOOLS & PIPELINE

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Technical artist working the seam between animation and engineering — five years inside AAA animation pipelines and two building the tools that feed them. I write animation exporters, editors and batch pipelines in C++ and Python against Unreal Engine and Maya, and I author the procedural animation and rig systems that consume them. I came up on the artist's side of the pipeline, calibrating volumes, cleaning takes, tracking faces and QA'ing retargets, so I build to how animators and riggers actually work.

EXPERIENCE

Crystal Dynamics — Senior Technical Artist I (prev. Technical Artist III)

Bellevue, Washington · August 2023 – May 2026

- Developed and maintained the core Animation Blueprint for a true-first-person shooter, holding view-model accuracy to the standard shooter gameplay clarity requires. [1]
- Led all LiveLink and motion capture preview operations.
- Integrated a procedural weapon sway system in Control Rig to add view-model variation, alongside a Control Rig multi-execution setup and a crowd facial emotion system. [1]
- Built and optimized MetaHuman animation pipelines supporting both stereo HMC and iPhone capture devices.
- Contributed the hero shot for the release trailer.

WB Games / Avalanche — Technical Artist (Motion Capture) (promoted from Motion Capture Tech)

Salt Lake City, Utah · March 2021 – August 2023

- Owned creation and maintenance of the motion capture pipeline: optimizations, hand-off formats, and getting data to its destination against a defined quality standard. [2]
- Cut body retargeting to next-day delivery by automating the hand-off steps between capture and animation.
- Baked Faceware project metadata into the retarget file so an entire project could be rebuilt from a script instead of by hand.
- Built an in-engine realtime preview that persists stage locations to file, letting animators re-establish a shoot's staging without the volume.
- Shipped a Python asset-optimization solution across multiple projects within six months.
- Ran day-to-day operation of a mocap volume on a very small team, full service prep to post — Vicon calibration, scheduling, staging, MotionBuilder-to-LiveLink setup, animation QA, retargeting, HMC setup, Faceware tracking, and MetaHuman Animator with Technoprops HMC.
- Scripted and staged a livestreamed motion capture demonstration to a 200,000-viewer audience. [2]

Earlier

- **House of Moves** — Stage and Ops Assist, Utah, 2022. Set up and tore down a 174-camera volume; ran MotionBuilder and worked directorial feedback on ideal framing.
- **Apple** — T2 Mac+ Advisor, 2016–2021. Last-line support for emerging OS defects: reproduced and documented fringe software cases for engineering.
- **University of Utah, College of Health** — Mocap Tech, 2015–2016. Motion capture for the Orthopedic Research Department on a force-plate volume; gait-analysis walkcycle cleanup in Nexus 2.2.
- **NOAA** — Technical Assistant, Honolulu, 2012–2013. Assisted the senior IT technician; built a redundant RAID 6 server and set data-destruction policy.

SELECTED PROJECTS

MHA Manager — Unreal Engine 5 C++ editor plugin [3]

Sole author — design, C++ implementation, documentation. Releasing 2026.

- Turns MetaHuman Animator from a manual one-take-at-a-time process into a validated batch queue across five job types, with auto asset discovery, parallel worker processes, per-job progress and live logging.

- Editor panel exposing 35+ per-job solve options through a dynamic column picker with inline cell editing; stereo calibration solved from checkerboard takes.
- Headless mode: a `-run=MHAManager` commandlet driven by a manifest path with distinct process exit codes, so a batch drops into CI and failed jobs retry without re-solving completed takes.
- MCP toolset registered against the engine's native `ModelContextProtocol` / `ToolsetRegistry` stack, degrading to a clean no-op where that stack is absent.
- Authored the full end-user documentation set, installation through troubleshooting.

Unreal Fellowship: Games — arcade project [4]

- Three-week light-gun arcade build in Unreal using Blueprints, Control Rig, retargeting and Sinden Lightgun hardware integration; wrote the public postmortem on the Unreal forums.

Shipped and released work

- **A Boy and His Beard** — released twin-stick beat 'em up on Steam. Production scheduling, mechanics and enemy design, balancing, main character creation, rigging, animation, and publishing. [5]
- **Play Two Heroes** — Overwatch Workshop mod with six coded systems and documented edge cases. Officially acknowledged by Blizzard; ranked 11th most-played by duration with the 2nd-highest average player retention. [6]
- **ThriveVR** — VR physical therapy prototype; Most Popular at Show, Games for Health 2016. [7]

EDUCATION & TRAINING

BS, Game Design — University of Utah, 2018. Additional Computer Science coursework taken beyond degree requirements.

Maya Plugin — C++ in Entertainment — Rigging Dojo, 2022. Maya's C++ API in Visual Studio: custom nodes and quality-of-life tooling.

Unreal Fellowship: Games — Epic Games, 2024.

Become a Technical Artist — ELVTR, 2024.

Associate's, New Media Arts (Digital Animation) — Kapi'olani Community College, 2012. Traditional arts foundation into 3D, rigging specialization.

TECHNICAL SKILLS

Languages C++ (Unreal editor plugins, Maya custom nodes), Python (pipeline automation, Maya/MotionBuilder tooling), Blueprints, MEL

Engines Unreal Engine 5 — Control Rig, Animation Blueprint, LiveLink, commandlets, editor UI, plugin build configuration, retargeting; Unity

DCC & capture Maya, MotionBuilder, Vicon (Shogun, Nexus), Faceware, MetaHuman Animator, Technoprops HMC, iPhone capture, Mudbox, Photoshop

Pipeline Animation exporters and editors, batch and headless automation, manifest-driven reproducible jobs, camera calibration, asset optimization, animation QA, Git, GitHub Actions, technical documentation

WORK SAMPLES

Bracketed numbers link to supporting media at hawaiianspawn.com — [1] /projects/perfect-dark, [2] /projects/hogwarts-legacy, [3] /marketplace/mha-manager (with full documentation beneath it), [4] /projects/escape-from-editor, [5] /projects/a-boy-and-his-beard, [6] /projects/play-two-heroes, [7] /projects/thrive-vr. Rigging and generalist reels at /projects/character-rig.